Network Exposure and Traffic Monitoring for Security Analysis

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1. ABSTRACT

With the rapid growth of networked systems and increasing dependence on internet-based services, ensuring system security and minimizing vulnerabilities has become a major concern in the field of cybersecurity. Many systems remain exposed to threats due to open ports, misconfigured services, and lack of proper monitoring mechanisms. These vulnerabilities can be easily exploited by attackers to gain unauthorized access, disrupt services, or compromise sensitive data. In many cases, users and administrators are not even aware of such exposures, which increases the risk of cyberattacks. Therefore, understanding how systems interact over a network and identifying potential weaknesses is essential for building secure environments.

This project focuses on developing a basic intrusion detection and response system by integrating port scanning, service monitoring, and log analysis techniques. Network scanning is performed using Nmap to identify open ports and active services on the system, which helps in assessing the level of exposure and identifying possible entry points for attackers. Based on the scanning results, automated shell scripts are used to take immediate defensive actions such as stopping vulnerable services. This automation reduces manual effort and ensures that risks are handled promptly, thereby improving system security and reliability.

In addition to port monitoring, the project also includes a Python-based log monitoring system that continuously analyzes Apache server logs in real time. The system detects suspicious activities such as repeated 404 errors and unauthorized attempts to access restricted paths like admin pages. These activities often indicate potential probing or attack attempts. Overall, the system highlights the importance of proactive monitoring, automation, and simple security practices in maintaining a secure computing environment.

2. PROBLEM STATEMENT

Modern computer systems are continuously connected to local and global networks, making them highly susceptible to various types of cyber threats. One of the major challenges in maintaining system security is the presence of open ports and running services that are not properly monitored or managed. These open ports can act as gateways for attackers to exploit vulnerabilities and gain unauthorized access to the system. In many cases, users are unaware of which ports are open or which services are actively running, leading to increased exposure and risk.

Another significant issue is the lack of visibility into system behavior and network communication patterns. Without proper tools and monitoring mechanisms, it becomes difficult to understand how data flows within the system and how external entities interact with it. This lack of awareness makes it challenging to detect abnormal activities or identify potential threats at an early stage. As a result, systems remain vulnerable to attacks that could have been prevented with proper monitoring and analysis.

Furthermore, system logs, which contain valuable information about system activities, are often underutilized or completely ignored. Logs can reveal important details about errors, access attempts, and unusual behavior, but without real-time monitoring, these insights are not effectively used. Suspicious activities such as repeated failed requests or unauthorized access attempts may go unnoticed until significant damage has already occurred. This project addresses these issues by providing a simple and automated solution that combines port scanning, service control, and log monitoring to improve system security and awareness.

3. OBJECTIVE

The primary objective of this project is to design and implement a simple yet effective system for monitoring network activity and detecting potential security threats. The system aims to identify open ports and active services using network scanning techniques, which helps in understanding the system's exposure and possible vulnerabilities. By regularly scanning the system, it becomes easier to detect unexpected changes in port status and take necessary actions to maintain security.

Another important objective of the project is to automate the process of detecting and responding to vulnerabilities. Instead of relying on manual monitoring, the system uses shell scripts to automatically check for open ports and take appropriate actions such as stopping services that may pose a security risk. This automation not only improves efficiency but also ensures that threats are handled quickly without delay, reducing the chances of exploitation.

In addition to port monitoring and automated response, the project also aims to analyze system logs in real time to detect suspicious activities. By continuously monitoring log files using a Python script, the system can identify abnormal patterns such as repeated errors or unauthorized access attempts. This helps in early detection of potential threats and allows users to take preventive measures. Overall, the project aims to provide a practical and beginner-friendly approach to understanding and implementing basic cybersecurity techniques.

4. SYSTEM DESIGN

The system is designed as a combination of multiple components that work together to monitor and secure the system effectively. The design includes three main modules: port scanning, service control, and log monitoring. Each module performs a specific function and contributes to the overall goal of detecting vulnerabilities and responding to potential threats. The integration of these modules ensures a structured and efficient approach to system security.

The port scanning module uses Nmap to identify open ports and running services on the system. This module plays a crucial role in detecting exposed services that may be vulnerable to attacks. By scanning the system regularly, it helps in identifying any unexpected changes in port status and ensures that all active services are accounted for. This information is essential for assessing the security posture of the system.

The service control module is responsible for taking action based on the results of the port scan. If a critical port is found open, the module automatically stops the associated service using predefined scripts. This reduces the risk of exploitation and ensures that unnecessary or vulnerable services are not left running. The log monitoring module complements this by continuously analyzing system logs to detect suspicious activities. Together, these components form a comprehensive system for monitoring and improving security.

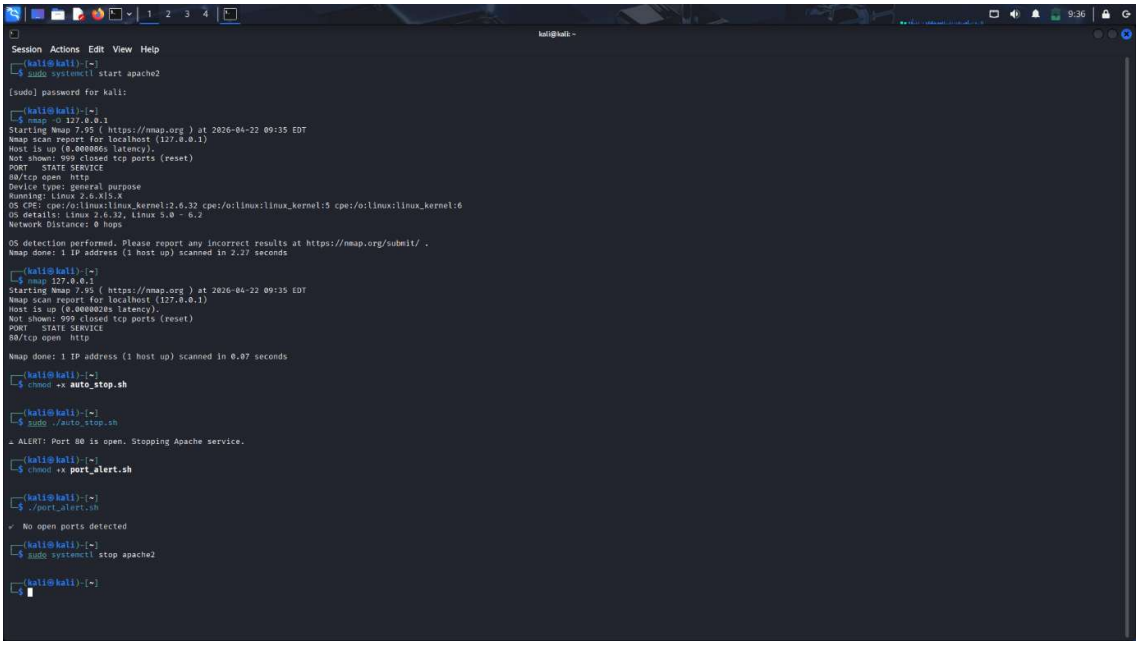
5. CONCLUSION

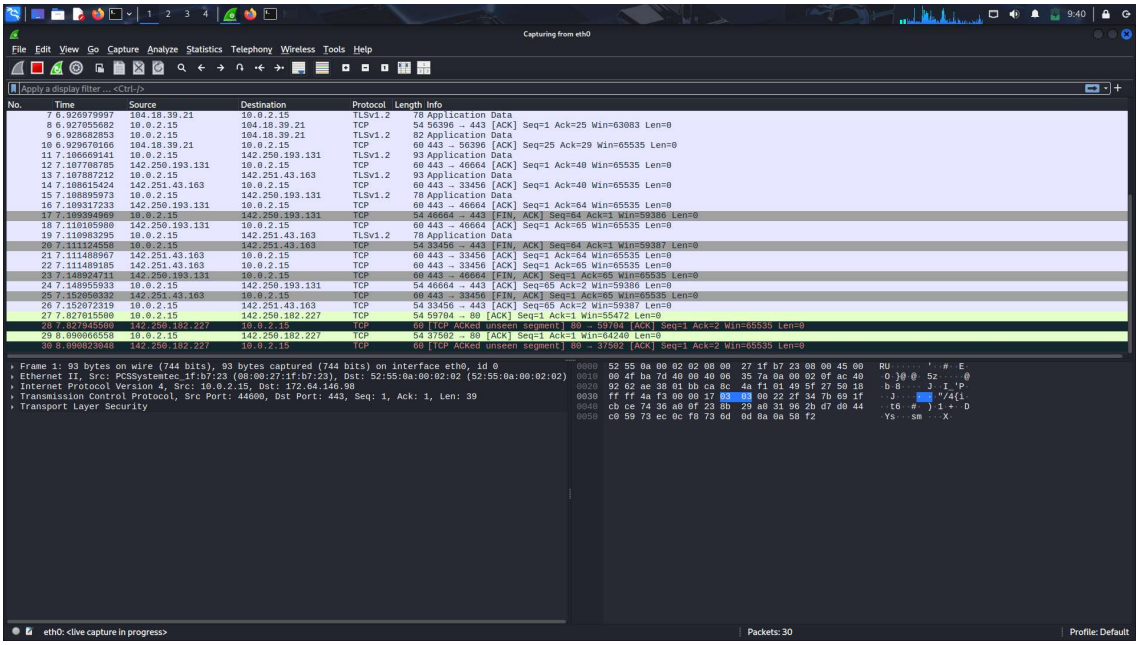
This project demonstrates a simple yet effective approach to improving system security through the integration of port scanning, automated service control, and log monitoring techniques. By identifying open ports and monitoring system activity, the system helps in detecting potential vulnerabilities and reducing the risk of unauthorized access. The use of widely available tools such as Nmap and Python makes the system accessible and easy to implement for beginners.

The automation of detection and response processes is a key feature of this project, as it reduces the need for manual intervention and ensures timely action. By automatically stopping vulnerable services and continuously monitoring system logs, the system provides a proactive approach to security. This highlights the importance of automation in modern cybersecurity practices and demonstrates how simple techniques can be used to build effective security solutions.

In conclusion, the project provides a strong foundation for understanding basic concepts in cybersecurity, including network scanning, intrusion detection, and system monitoring. It emphasizes the importance of continuous monitoring, quick response, and proactive defense in maintaining a secure environment. The system can be further enhanced with advanced features such as alert notifications and integration with other security tools, making it a valuable starting point for more complex security implementations.

6. SCREENSHOTS





~/log_monitor.py - Mousepad

File Edit Search View Document Help

```python
import time

log_file = "/var/log/apache2/access.log"

print("Monitoring Apache logs...")

with open(log_file, "r") as file:
    file.seek(0, 2)  # Move to end of file

    while True:
        line = file.readline()
        if not line:
            time.sleep(1)
            continue

        if "404" in line:
            print("⚠️ Suspicious Activity: 404 Error detected!")
            print(line.strip())

        if "admin" in line:
            print("⚠️ Suspicious Path Access Detected!")
            print(line.strip())
```

Shell No. 1

Session Actions Edit View Help

```bash
#!/bin/bash

TARGET="192.0.0.1"
OPEN_PORTS=$(nmap -p 1-1024 $TARGET | grep open)

if [ -n "$OPEN_PORTS" ]; then
    echo "⚠ Alert: Open ports detected!"
    echo "$OPEN_PORTS"
else
    echo "✓  No open ports detected"
fi
```

1,1 All

Shell No. 1

Session Actions Edit View Help

```bash
#!/bin/bash

TARGET="127.0.0.1"
PORT=80

RESULT=$(nmap -p $PORT $TARGET | grep "open")

if [ -n "$RESULT" ]; then
    echo "⚠ ALERT: Port $PORT is open. Stopping Apache service."
    sudo systemctl stop apache2
else
    echo "✅ Port $PORT is closed. No action needed."
fi
```

"~/auto_stop.sh" 13L, 279B 1,11 All